

C4 Database and Basic SQL

C4-1a: Introduction to Database

Database Concept

What is Database?

A database is a collection of related data, stored in an organised or logical manner.

Storing data in a database allows us to access and manage the data. Examples of databases in real-life include:

- Medical records of patients in a hospital
- Supermarket inventory
- Contact list

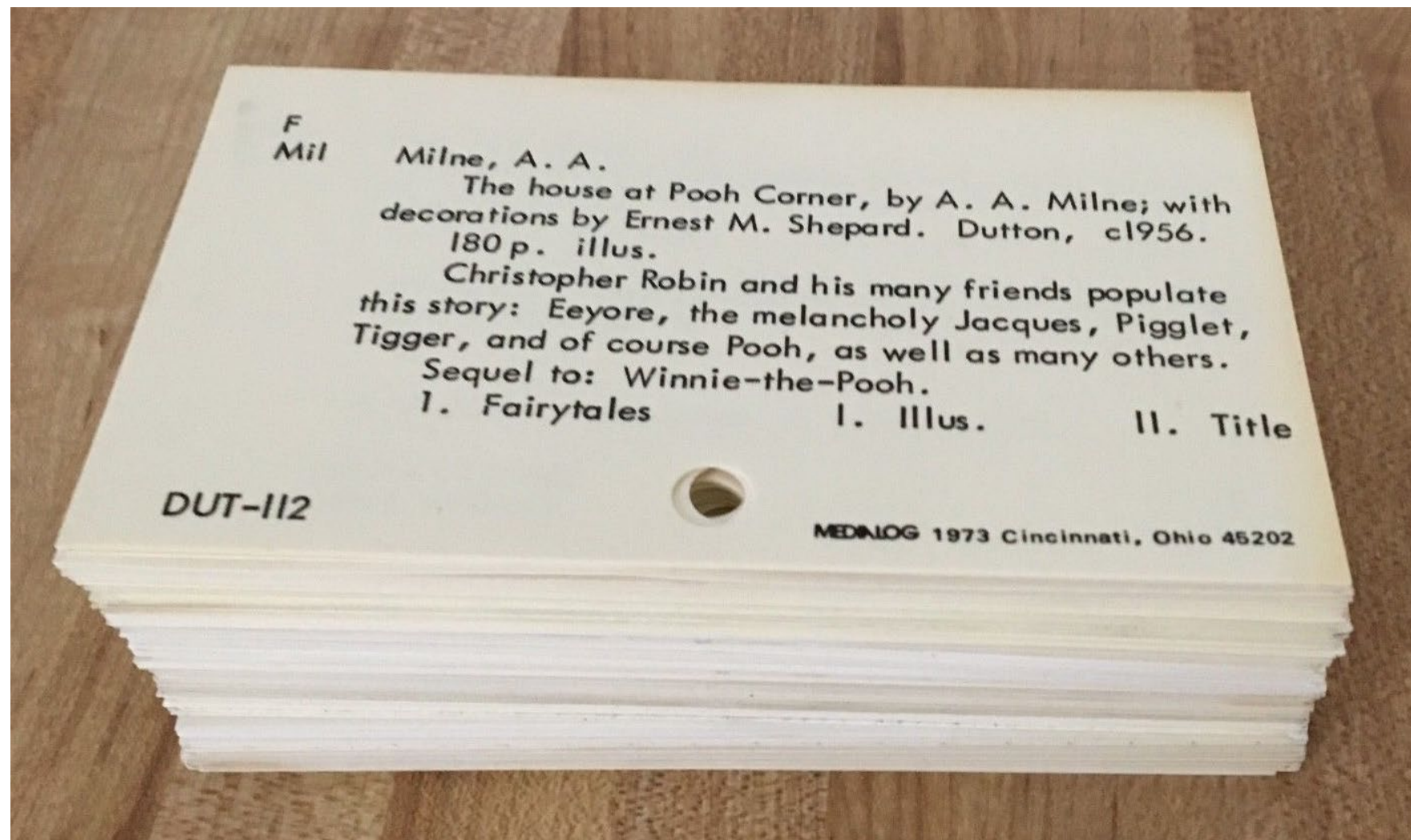
Old School – Library Card Catalog



<https://www.youtube.com/watch?v=wvcatnesYcw&t=23s>

Old School – Library Card Catalog

- The index cards



Physical to Digital

Card Index System

- The cards would have a header line and the cards would then be filed in some determined order, for example, customer name filed in alphabetical order.
- That was fine if we wanted to find a particular customer by name but what if we wanted to find the customer living at “13 The High Street” or the customer with the contact number ‘91234567’?
- The major limitation of a manual card index is that the cards can only be held in one particular order and so can only be searched on one data item.

Digital solution to store and manage large records

C4-1b: Database, Tables, Fields, Records

Database Concept

Database, Tables, Fields and Records

- A **database** is an organized collection of data, typically structured into **tables** to facilitate efficient access, management, and updating of information.
- Each **table** within a database consists of rows, known as **records**, and columns, known as **fields**.
- A **record** represents a single entry in the table, containing data related to a specific item or entity.
- Each **field** within a record holds a particular type of data, such as text, numbers, or dates, corresponding to a specific attribute of the entity.
- This tabular structure allows for the clear organization of data, making it easier to query, analyze, and maintain.

Database, Tables, Fields and Records

Each **table** within a database consists of rows, known as **records**, and columns, known as **fields**.

There are 10 records in this table,

- each records contain data for the 4 fields:
- RegNo
 - Name
 - Gender
 - MobileNo

Field names are "single word", e.g. **MobileNo**, not "Mobile No".

RegNo	Name	Gender	MobileNo
1	Adam	M	92313291
2	Adrian	M	92585955
3	Agnes	F	83324112
4	Aisha	F	88851896
5	Ajay	M	94191061
6	Alex	M	98675171
7	Alice	F	95029176
8	Amy	F	98640883
9	Andrew	M	95172444
10	Andy	M	95888639

Table Description

A table description is expressed as:

TableName (Attribute1, Attribute2, Attribute3, ...)

RegNo	Name	Gender	MobileNo
1	Adam	M	92313291
2	Adrian	M	92585955
3	Agnes	F	83324112
4	Aisha	F	88851896
5	Ajay	M	94191061
6	Alex	M	98675171
7	Alice	F	95029176
8	Amy	F	98640883
9	Andrew	M	95172444
10	Andy	M	95888639

The data value for each field should be **atomic**.

- Each record can only have one mobile number.

Each record (row) should be **unique**, same record cannot appear twice within a table.

The **order of the fields** (column) is not important.

Each field must be **unique**.

STUDENTS (RegNo, Name, Gender, MobileNo)

Primary Key (PK)

A primary key is a **field** that helps to uniquely identifies a particular record in a table; the data value of this field should not change over time.

Some good choices:

{RegNo, Name, Gender, MobileNo},

{RegNo, Name, MobileNo},

{Name, MobileNo}, {RegNo, Name},

{RegNo, Gender}, **{RegNo}** ..

Which one is the best choice to be the PK?

RegNo	Name	Gender	MobileNo
1	Adam	M	92313291
2	Adrian	M	92585955
3	Agnes	F	83324112
4	Aisha	F	88851896
5	Ajay	M	94191061
6	Alex	M	98675171
7	Alice	F	95029176
8	Amy	F	98640883
9	Andrew	M	95172444
10	Andy	M	95888639

Wrong choices are:

{Gender}, {Name}, {MobileNo}

Composite Primary Key (PK)

A composite primary key is a **collection of fields** that helps to **uniquely identifies** a particular record in a table.

Some possible choices are:

{Name, Gender, MobileNo},

{Name, MobileNo}

Which one is a better choice to be the PK?

RegNo	Name	Gender	MobileNo
1	Adam	M	92313291
2	Adrian	M	92585955
3	Agnes	F	83324112
4	Aisha	F	88851896
5	Ajay	M	94191061
6	Alex	M	98675171
7	Alice	F	95029176
8	Amy	F	98640883
9	Andrew	M	95172444
10	Andy	M	95888639

Wrong choices are:

{Name, Gender},

{Gender, MobileNo}

{Name}, {Gender}, {MobileNo}